

Explore
Learn
Build
Create



LDR SENSOR

Senses Light Around You!

Brighter Ideas, Brighter Tomorrow!



Build
Smart
Solutions
for a
Brighter
World!



What is an LDR Sensor?

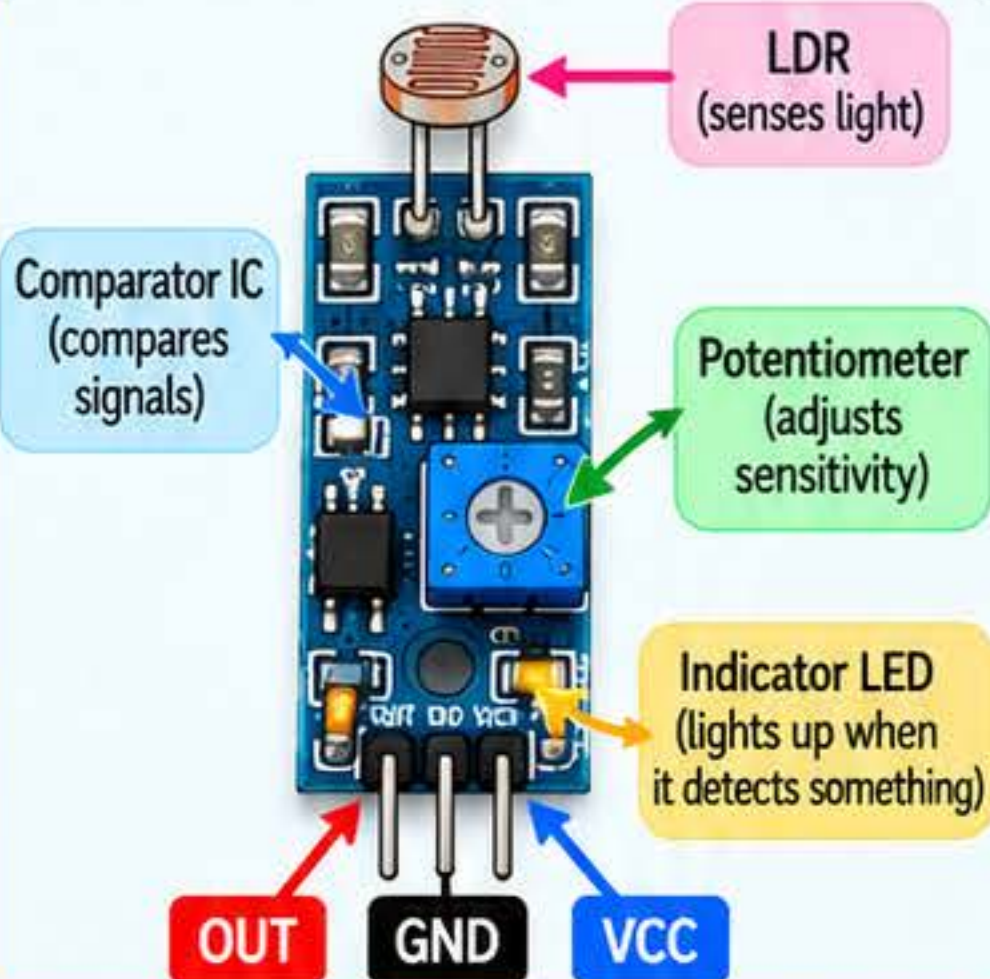
An LDR (Light Dependent Resistor) is a sensor that changes its resistance based on the amount of light falling on it.



It senses light,
just like our eyes!

It has high resistance
in the dark and low
resistance in bright light.

Parts of an LDR Sensor Module



How Does It Work?



High Resistance
(less current)

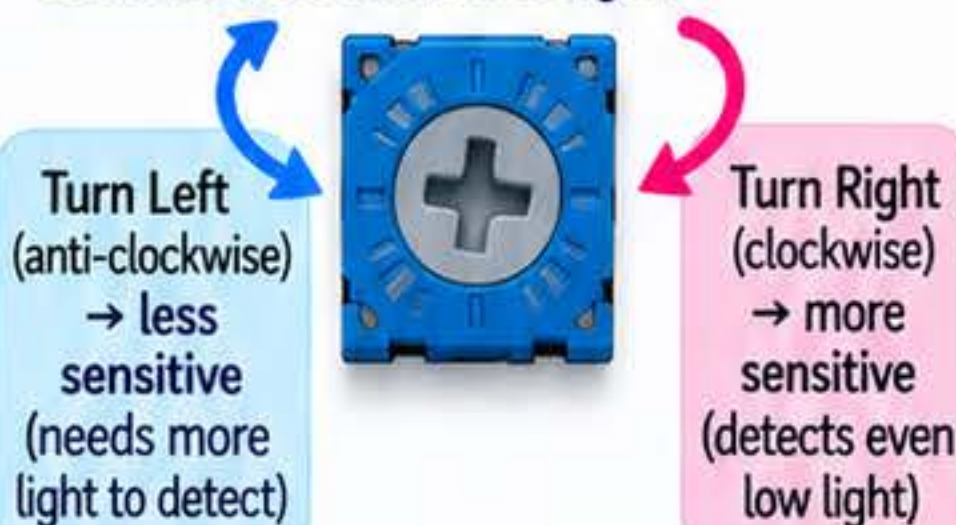


Low Resistance
(more current)

When light falls on the LDR, its resistance changes. The module converts this change into a signal that can be read by a microcontroller like Arduino.

The Potentiometer

The potentiometer lets you adjust how sensitive the sensor is to light.



You can adjust it to make the module work best for your project!

What Can It Do?

An LDR sensor can be used to:

Detect Light and Dark



It can tell the difference
between day and night.

Automatic Lights



Can turn on lights
automatically when
it gets dark.

Measure Light Levels



Can be used to
measure how bright
or dark it is.

Smart Projects



Useful in many projects
like automatic street
lights, smart home
systems, and more!

Real-Life Examples



Street
Lights



Garden
Lights



Camera
Light Meters



Mobile Phones
(adjust screen
brightness)

Fun Fact!



LDRs help save
energy by turning
lights on only
when needed.
That's good for
our planet!



Small Sensors. Big Ideas!

Learn • Experiment • Create • A Brighter Future!

